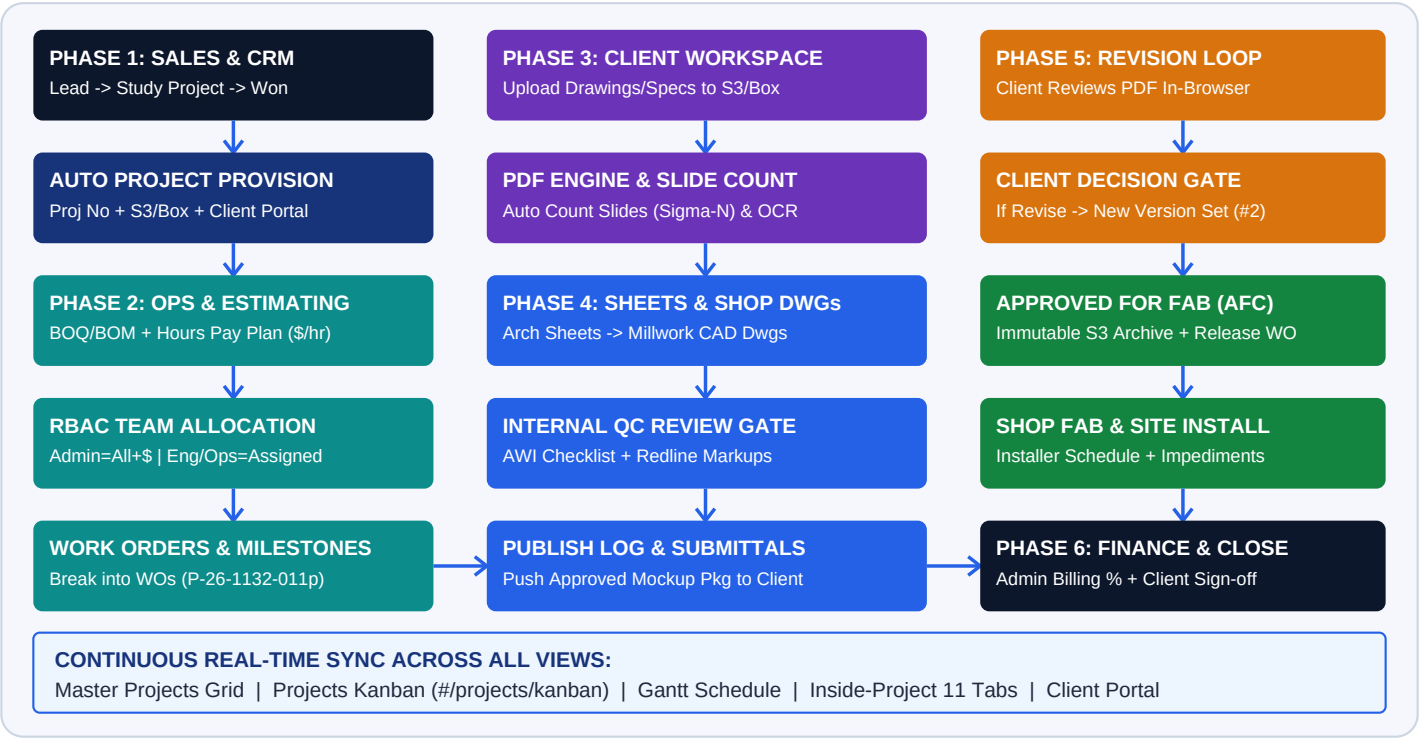
QUADECONNECT · COMPLETE WORKFLOW SPECIFICATION

QuadeConnect Project Management

ClickUp Work Management UX + Innergy Architectural Millwork ERP Domain

100% Connected Flow: CRM Intake -> Estimating & Hours Pay -> Master Grid & Kanban -> Inside-Project
11 Millwork Tabs -> S3/Box Multi-Slide PDF Viewer -> QC & Submittal Loop (REVX) -> Billing & Closeout

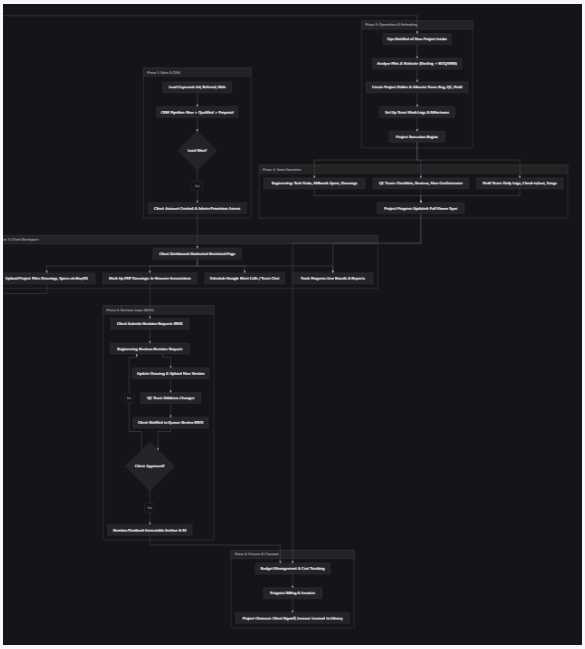
SEC 01 Master Visual Process Flowchart (6 Connected Lifecycle Phases)



SEC 02 Document Roadmap · What Every Section of This PDF Covers

Pages 1-2: Executive Architecture, Your Uploaded 6-Phase Flow Diagram & Role-Based Cost Masking (Admin vs. Engineers/Ops)
Pages 3-4: Line-by-Line Process Connection (Phases 1 to 6: CRM, Estimating, Hours Pay Plan, S3/Box Intake, Execution, REVX)
Page 5: Master Projects Hub · Grid View (Ref Image 1) & ClickUp/Innergy Kanban Board (app.innergy.com/#/projects/kanban)
Page 6: Inside a Project · Deep Breakdown of All 11 Architectural Millwork Tabs (Details, Study Projects, Work Orders, etc.)
Pages 7-8: Sheets Module Deep Dive (Ref Images 4 & 5) · Grid, Multi-Slide PDF Viewer, Slide Counts (Sigma-N) & Revision History
Page 9: Shop Drawings Module Deep Dive (Ref Image 3) · Mockup Packages, Work Order Links & Publish Log Gatekeeper
Pages 10-12: Work Orders, Study Projects, QC Review, Submittals, S3/Box Pipeline, Database ER Model & Feature Comparison Matrix

SEC 03 Your Uploaded 6-Phase Flow Diagram & System Mapping



Reference Image 2: Your Uploaded 6-Phase Flow Diagram (Sales/CRM -> Ops/Estimating -> Client Workspace -> Execution -> REVX -> Finance)

SEC 04 Role-Based Access Control (RBAC) & Strict Cost/Pay Visibility Rules

As required for QuadeConnect: Admins have unrestricted access to all projects, contract financials, billing percentages, and hourly pay rates. Engineers, Drafters, QC Reviewers, and Operations Heads are strictly limited to the specific set of projects they work on, and ALL cost/contract dollar values are stripped at the API level.

User Role	Project Access Scope	Contract & Cost Visibility	Hours Pay & Labor Plan	Allowed Modules & Actions
Super Admin / Admin	ALL Projects (100% Portfolio)	VISIBLE: Original Contract, Current Contract, WO Value, Billed $, Billed %, Est. Grand	VISIBLE: Sets $/hr pay rates per trade & views total labor $	Create Projects, Manage Budgets, S3/Box Config, Rate Cards, All 11 Tabs
Operations Head	Assigned Set of Projects Only	HIDDEN: Zero dollar or billing columns shown in Grid or Header	HOURS ONLY: Views budgeted hours vs. actual logged hours	Allocate Team, Manage Study Projects, Work Orders, Installer Schedule, QC
Project Manager	Assigned Set of Projects Only	HIDDEN: Sees only non-dollar badge ('Budget is up to date')	HOURS ONLY: Tracks engineering & shop task hours vs plan	Manage Details, Sheets, Shop Drawings, Submittals, Publish Log, RFIs
Engineer / Drafter	Assigned Set of Projects Only	HIDDEN: Stripped by backend API middleware	OWN HOURS: Logs hours on assigned drawings & study projects	Read Sheets, Upload Shop Drawings, Link Work Orders, Fix QC Redlines
QC Reviewer	Assigned Set of Projects Only	HIDDEN: Stripped by backend API middleware	OWN HOURS: Logs QC inspection hours	Open QC Queue, Run Millwork Checklists, Add PDF Redlines, Approve/Reject
Field / Installer	Assigned Set of Projects Only	HIDDEN: Stripped by backend API middleware	OWN HOURS: Daily check-in/out & site logs	View Installer Schedule, Read Published Drawings, Log Site Impediments
Client (External)	Own Project Portal Only	HIDDEN: Sees only their scope (unless Invoice Portal enabled)	HIDDEN: Never sees internal hours or rates	Upload Intake Files (S3/Box), View Published PDFs, Markup, Approve REVX

HOW BACKEND API ENFORCES COST HIDING & PROJECT SCOPING:

1. Query Scoping: SELECT * FROM projects WHERE id IN (SELECT project_id FROM project_assignments WHERE user_id = current_user.id)
2. DTO Serializer Guard: If user.role != 'ADMIN', strip [original_contract, current_contract, wo_value, billed, billed_pct, est_cost, hourly_rate].

PHASE 1 Sales & CRM Pipeline -> Pre-Bid Study Project -> Project Creation

STEP 1.1 Lead Capture & Millwork Bid Intake

- Incoming bid opportunities arrive via General Contractors (Whiting Turner, Hensel Phelps, Beck Group), Architects, Referrals, or Website.
- Captures Customer Name, Project Location, Bid Due Date, Architectural Specification Sections (e.g., CSI Division 06 40 00 Architectural Woodwork), and preliminary drawing links.

STEP 1.2 CRM Pipeline Progression & Study Project Trigger

- Deal progresses through CRM stages: New Lead -> Qualified -> Study Project / Estimating -> Proposal Sent -> Negotiation.
- For complex custom millwork (curved reception desks, acoustic wall panels, exotic veneers), estimator triggers a 'Study Project' to analyze joinery, CNC feasibility, and material waste factors.

STEP 1.3 Decision Gate: Lead Won? -> Automated Project Provisioning

- If LOST: Deal archived with loss reason code for historical estimating analytics.
- If WON: System automatically creates a Master Project Record with sequential Project No. (e.g., P-26-1137), provisions the AWS S3 / Box cloud folder tree, creates the restricted Client Portal account, and notifies Operations Head.

PHASE 2 Operations & Estimating Setup (BOQ/BOM, Hours Pay Plan & Team Allocation)

STEP 2.1 Operations Intake, BOQ / BOM Breakdown & Hours Pay Plan

- Operations Head & Estimator review intake files and generate the Bill of Quantities (BOQ) and Bill of Materials (BOM: substrates, veneers, laminates, solid surface, hardware).
- Admin configures the Hours Pay Plan: sets hourly pay rates ($/hr) per department (Engineering/Drafting, CNC Machining, Assembly/Cabinetmaking, Finishing, Site Installation) and assigns budgeted hours per phase.
- System computes Estimated Grand Total Cost = (Budgeted Hours x Hourly Rate) + Material BOM Cost + Subcontractor/Overhead allocation (visible strictly to Admin).

STEP 2.2 Project Folder Setup & Role-Based Team Allocation

- S3 / Box auto-creates subfolders: /Sheets, /Shop-Drawings, /Work-Orders, /Submittals, /QC-Reviews, /Study-Projects, /Files, and /Client-Uploads.
- Operations Head assigns the dedicated Project Manager, Lead Engineers/Drafters, QC Reviewer, and Field Superintendent.
- RBAC immediately grants those assigned users access to this project in their Grid and Kanban views·while hiding all contract and cost columns.

STEP 2.3 Work Order Breakdown & Milestone Scheduling

- Project is broken down into buildable Work Orders (e.g., P-26-1132-011p Table of Contents, P-26-1132-025p Restroom Mockup, P-26-1132-022p Bar Counter).
- ClickUp-style task dependencies and milestones are linked: Sheets Review -> Shop Drawing Drafting -> QC Review -> Client Submittal -> CNC Fabrication -> Site Installation.

PHASE 3 Client Workspace & Cloud Document Intake (AWS S3 / Box)

STEP 3.1 Restricted Client Portal & Direct S3 / Box File Uploads

- Client logs into their dedicated workspace and uploads architectural drawing sets (IFP, Bulletins, Addenda), finish schedules, and hardware cutsheets.
- Files upload via pre-signed URLs directly to AWS S3 or Box, automatically linking to the project's Files and Sheets tabs and notifying the PM.

STEP 3.2 In-Browser PDF Opening, Markup & Live Progress Tracking

- All linked S3/Box files open directly inside the QuadeConnect web application (zero local software required) with tools to mark up redlines, schedule Google Meet calls, and track live Kanban progress.

PHASE 4 Team Execution Inside the Project (Engineering, QC & Field Teams)

STEP 4.1 Engineering Execution: Architectural Sheets -> Millwork Shop Drawings

- Engineers open the project's Sheets tab to study contract drawings (e.g., G0.02 Project Information, ADA Standards, Floor Plans) and field dimensions.
- Engineers produce detailed CAD Shop Drawings (elevations, sections, joinery details, panel layouts, hardware callouts) and upload PDFs to the Shop Drawings tab linked to specific Work Orders.
- Automated PDF Processor parses every uploaded PDF: extracts total slide/page count (displayed as Sigma-9, Sigma-4, Sigma-2), renders high-res page thumbnails, and logs the Version Set.

STEP 4.2 Internal QC Review Gate (Checklist + Redline Verification)

- Uploaded Shop Drawings enter 'Pending QC Review'. The QC Reviewer opens the drawing in the split-pane viewer and verifies it against the Millwork QC Checklist (dimensions, AWI grade, veneer grain matching, edge banding, hardware clearances, blocking).
- If errors exist, QC draws non-destructive redlines directly on the PDF slide and marks status 'QC Rejected' -> Engineer is notified to upload a corrected file.
- If passed, QC marks 'QC Approved' -> Drawing moves to the Publish Log queue.

STEP 4.3 Field Team Execution: Installer Schedule, Daily Logs & Impediments

- Field crews view the Installer Schedule tab, log daily check-in/out hours, upload site readiness photos, and log any site blockers in the Impediments tab (e.g., 'GC drywall not ready on Level 3').
- All engineering, QC, and field updates synchronize in real time with the Master Project Grid, Kanban Board, and Client Progress Tracker.

PHASE 5 Revision Loop (REVX), Version History & Submittal Sign-Off

STEP 5.1 Publish Log -> Client Submittal & Revision Request (REVX)

- PM selects QC-approved Shop Drawings in the Publish Log (e.g., PUBLISH LOG 0/18) and issues a formal Submittal package (e.g., MOCKUP PACKAGE #1) to the Client/Architect.
- Client reviews the multi-slide PDF in-browser. If changes are needed, the client pins redline annotations and submits a Revision Request (REVX).

STEP 5.2 Engineering Version Update -> QC Validation -> Immutable Archive

- Engineer reviews the client's REVX redlines, updates the CAD drawing, and uploads a new Version Set (e.g., MOCKUP PACKAGE #2).
- The previous version remains permanently accessible in the bottom-left Revision History panel (with side-by-side diff comparison and slide counts).
- QC validates the revised drawing -> Client re-reviews and clicks Approve -> System locks the PDF as an Immutable Archive in S3/Box and releases the linked Work Order for CNC & Shop Fabrication.

PHASE 6 Finance, Progress Billing & Project Closeout (Admin Only)

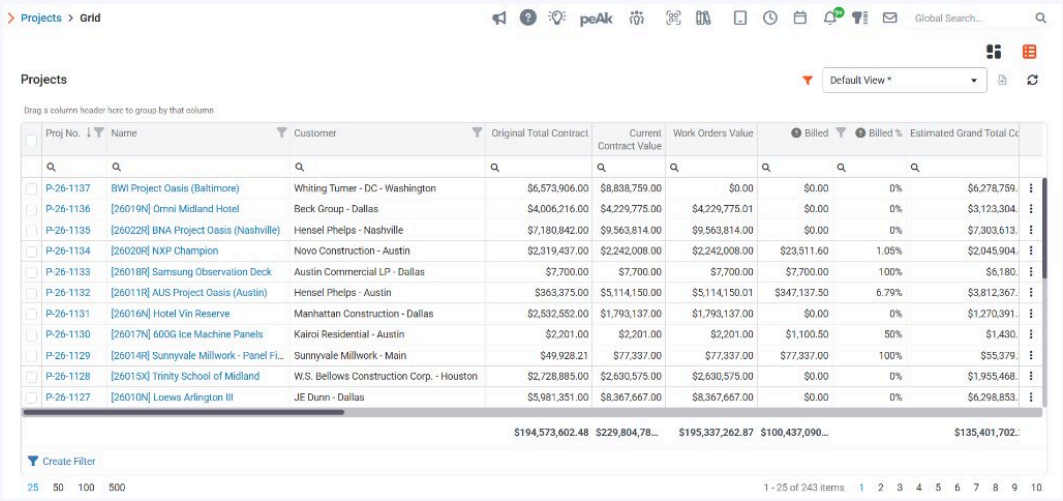
STEP 6.1 Real-Time Budget vs. Actual Cost Tracking (Admin View)

- Admin monitors Original Total Contract, Approved Change Orders, Current Contract Value, Work Orders Value, and Estimated Grand Total Cost (actual logged hours x hourly pay rate + material POs).

STEP 6.2 Progress Billing (Billed $ & Billed %) & Closeout Archive

- Generates AIA-style progress billing invoices, updates Billed ($) and Billed % in the Master Grid, captures final Client Sign-Off, and archives Lessons Learned.

SEC 05 Master Projects Grid View (Ref: Image 1 · Admin Full Financial View)



Reference Image 1: Innergy Master Projects Grid View · Proj No, Name, Customer, Contract Values, Work Orders Value, Billed $ & %

Column-by-Column Specification of the Master Projects Grid

1. **Proj No. (Sortable / Filterable):** Unique project identifier (e.g., P-26-1137, P-26-1136, P-26-1132). Clicking opens the 11-tab Project Hub.
2. **Name (Searchable):** Includes internal job code + descriptive title (e.g., [26011R] AUS Project Oasis (Austin), [26019N] Omni Midland
3. **Customer (Groupable):** General Contractor or Owner (e.g., Whiting Turner - DC, Beck Group - Dallas, Hensel Phelps - Austin). Drag hea
4. **Original Total Contract [ADMIN ONLY]:** Initial signed base contract amount (e.g., $6,573,906.00; portfolio footer sum = $194,573,602.48).
5. **Current Contract Value [ADMIN ONLY]:** Original Contract + Approved Change Orders (e.g., $8,838,759.00; portfolio footer sum = $229,804,78...).
6. **Work Orders Value [ADMIN ONLY]:** Cumulative dollar value of issued production Work Orders (e.g., $5,114,150.01; portfolio sum = $195,337,262.87
7. **Billed & Billed % [ADMIN ONLY]:** Total invoiced to date ($347,137.50) and percentage billed against Current Contract (6.79%, 50%, 100%).
8. **Estimated Grand Total Cost [ADMIN ONLY]:** Projected total cost combining Labor (Hours x Pay Rate) + Materials + Overhead ($3,812,367.00).

SEC 06 Projects Kanban View (Ref: app.innergy.com/#/projects/kanban + ClickUp UX)

Clicking the Kanban icon in the top-right switcher transforms the Projects Grid into an interactive ClickUp + Innergy Kanban Board (#/projects/kanban). Projects move across architectural millwork lifecycle columns via drag-and-drop:

1. STUDY / PRE-BID	2. ENGINEERING	3. QC & SUBMITTAL	4. SHOP PRODUCTION	5. SITE INSTALL
P-26-1140 Lobby Veneer yield study 2 Study Items	**P-26-1137 BWI Oasis** 216 Sheets | 52 Dwgs PM: R. Redmond	**P-26-1132 AUS Oasis** Mockup Pkg #5 Out 42 Submittals	**P-26-1135 BNA Oasis** 199 Work Orders CNC & Cabinet Assy	**P-26-1133 Samsung** Installer Crew A 100% Billed / Close
+ Drag Card Here WIP Limit & Filters	+ Drag Card Here WIP Limit & Filters	+ Drag Card Here WIP Limit & Filters	+ Drag Card Here WIP Limit & Filters	+ Drag Card Here WIP Limit & Filters

Kanban Capabilities: Swimlanes by Customer / PM / Priority | Card Counters (Sheets, Shop Drawings, WOs, Impediments) | Cost Badges Hidden for Engine

SEC 07 Inside-Project Persistent Header & Role-Based Financial Pills

Projects > [26011R] AUS Project Oasis (Austin) > Shop Drawings > Grid

P-26-1132 - [26011R] AUS Project Oasis (Austin)

Budget is up to date

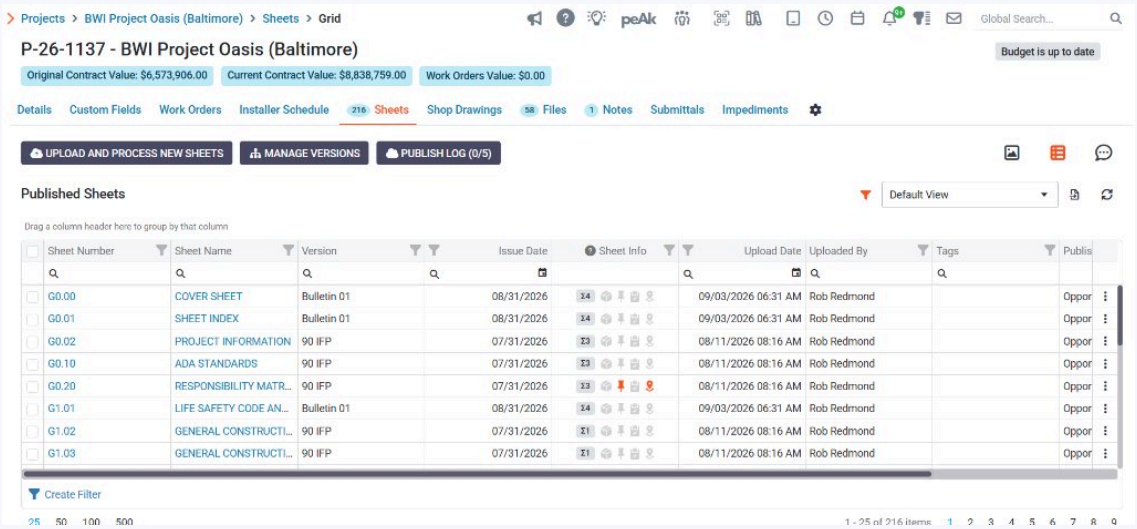
Original Contract Value: $363,375.00 | **Current Contract Value: $5,114,150.00** | **Work Orders Value: $5,114,150.01**

Details | Custom Fields | Study Projects | 199 Work Orders | Installer Schedule | 306 Sheets | 52 Shop Drawings | 174 Files | QC Review | 42 Submittals | 7 Impediments

SEC 08 Deep Functional Breakdown of All 11 Inside-Project Category Tabs

1. Details Tab Project Overview, Contacts	Stores Project No, Name, Customer, Architect, GC, Site Address, Contract Dates, Assigned Team (PM, Engineers, QC, Installers), and the Hours Pay Budget (Engineering hrs, CNC/Fab hrs, Finishing hrs, Install hrs). Admin sees full financial margin
2. Custom Fields Tab Dynamic Architectural Mill	ClickUp-style customizable fields: AWI Quality Grade (Premium/Custom/Economy), Wood Veneer Species & Cut (Plain Sliced White Oak, Rift Walnut), Finish System (Conversion Varnish, Stain Match), Substrate (MDF, Particleboard, Fire-Rated), Hardware Brand
3. Study Projects Tab Pre-Engineering R&D, Mocku	Dedicated space for complex millwork engineering studies before releasing production drawings: 3D solid modeling trials, custom knife/molding profile studies, curved reception desk framing tests, veneer flitch yield studies, and value-engineering
4. Work Orders Tab (199) Buildable Production Scope	Divides the contract into numbered Work Orders (e.g., P-26-1132-011p, 025p, 022p). Tracks WO status (Draft -> Engineering -> Approved -> In CNC/Fab -> Shipped -> Installed), linked Shop Drawings, BOM material status, and budgeted vs. actual shop
5. Installer Schedule Tab Field Delivery Staging, Cr	Interactive Gantt and Calendar timeline for field installation teams. Tracks floor-by-floor delivery dates, assigned installation crews, daily site check-in/out logs, field measurements, and final punch-list completion.
6. Sheets Tab (306 / 216) Client / Architect Contrac	Holds all architectural, structural, and interior design sheets (G0.00, G0.02, A1.01). Features 'UPLOAD AND PROCESS NEW SHEETS', 'MANAGE VERSIONS' (90 IFP, Bulletin 01), 'PUBLISH LOG', automatic PDF slide counting (Sigma-4, Sigma-3), and split-pane
7. Shop Drawings Tab (52) Internal Millwork CAD Fabr	Holds millwork engineering drawings (100.10 Table of Contents, 101.00 Restroom Mockup, 102.49 Bar Counter) grouped by Mockup Package versions (#1..#5) and linked directly to Work Orders with multi-slide PDF viewing (Sigma-9, Sigma-2).
8. Files Tab (174 / 58) AWS S3 / Box Cloud Documen	Structured cloud folders for Project Specifications, Signed Contracts, Hardware Schedules, Finish Sample Photos, Field Verification Photos, and Client Uploads. Every file opens in-browser via secure S3/Box signed URLs.
9. QC Review Tab Internal Quality Control G	Central queue for QC Reviewers to inspect drafted Shop Drawings and physical shop mockups. Includes standardized AWI/Millwork checklists, non-destructive PDF redline tools, rejection/resubmission tracking, and pass/fail audit logs.
10. Submittals Tab (42) Formal Architect / GC Subm	Packages approved Shop Drawings, Product Data cutsheets, and physical veneer/laminate samples into numbered Submittals with automated transmittal cover sheets, Ball-in-Court tracking, and Architect approval stamps.
11. Notes (11) & Impediments Contextual Project Threads	Tracks active impediments delaying progress (e.g., 'Missing field dimensions in Lobby', 'Architect finish selection pending', 'GC site elevator unavailable') with priority, owner, and days-blocked counter.

SEC 09 Published Sheets Grid View & Automated PDF Processing (Ref: Image 5)



Reference Image 5: Published Sheets Grid (P-26-1137 BWI Project Oasis) · 216 Sheets, Version Sets & Slide Counts (Sigma-4, Sigma-3, Sigma-1)

Step-by-Step Workflow Inside the Sheets Grid

STEP 1 Upload & Process New Sheets (S3 / Box + OCR Title Block Scan)

- Clicking [UPLOAD AND PROCESS NEW SHEETS] opens the drag-and-drop uploader connected to AWS S3 / Box.
- When an architectural PDF package is uploaded, the backend PDF worker automatically counts the slides/pages inside each PDF, generates 300-DPI slide thumbnails, and runs OCR on the title block to extract Sheet Number (G0.00, G0.01, G0.02, G0.10, G0.20, G1.01) and Sheet Name (COVER SHEET, SHEET INDEX, PROJECT INFORMATION, ADA STANDARDS, RESPONSIBILITY MATRIX).

STEP 2 Understanding the 'Sheet Info' Column Badges (Slide Counts & Icons)

- Sigma Badge (Sigma-4, Sigma-3, Sigma-1): Shows exact number of slides/pages inside that sheet's PDF version history so engineers immediately know how many slides exist.
- 3D Cube Icon: Indicates linked 3D CAD / BIM reference model.
- Pushpin Icon (Orange when active, e.g., on G0.20): Indicates active pinned markups, RFIs, or callouts placed on that sheet.
- Document / Location Pin Icons: Shows linked Work Orders, Submittals, or floor-plan location markers.

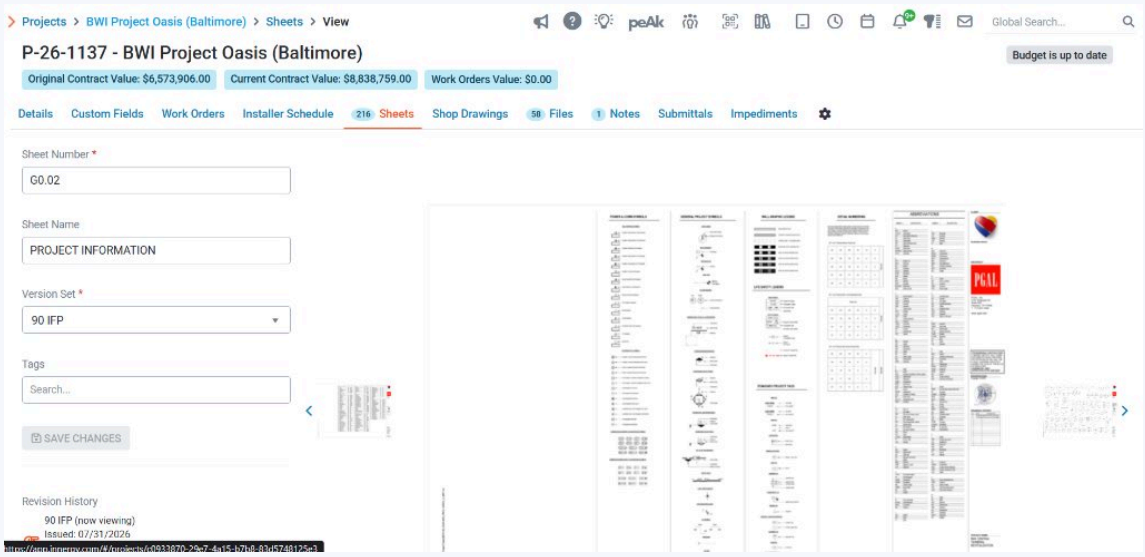
STEP 3 Manage Versions (90 IFP, Bulletin 01, Bulletin 02, 100% CD)

- Clicking [MANAGE VERSIONS] allows the PM to create and organize architectural issuance sets (e.g., '90 IFP' issued 07/31/2026 vs. 'Bulletin 01' issued 08/31/2026).
- When a revised sheet (like G0.00 or G1.01 in Bulletin 01) is uploaded with the same Sheet Number, QuadeConnect stacks it on top of the previous version ('90 IFP') and increments the Revision History counter.

STEP 4 Publish Log (0/5) · Controlled Release to Team & Client

- Newly processed sheets remain in staging until the PM reviews and clicks [PUBLISH LOG (0/5)], ensuring field crews and engineers only build from verified architectural issuances.

SEC 10 Inside Sheet View · Metadata, Revision History & Slide Viewer (Ref: Image 4)



Reference Image 4: Inside Sheet View (G0.02 Project Information) · Left Metadata & Revision History + Right Multi-Slide PDF Canvas

Detailed Breakdown of the Split-Pane Sheet Viewer (Line-by-Line from Image 4)

LEFT PANEL Sheet Metadata Editor & Version Set Selector

- Sheet Number * [G0.02]: Editable text field for AIA sheet code.
- Sheet Name [PROJECT INFORMATION]: Editable descriptive sheet title.
- Version Set * [90 IFP]: Dropdown selector allowing instant switching between '90 IFP', 'Bulletin 01', '100% CD', or 'ASI-01'.
- Tags [Search...]: Add custom filter tags (e.g., 'Millwork Scope', 'Lobby', 'ADA', 'Restrooms') and click [SAVE CHANGES].

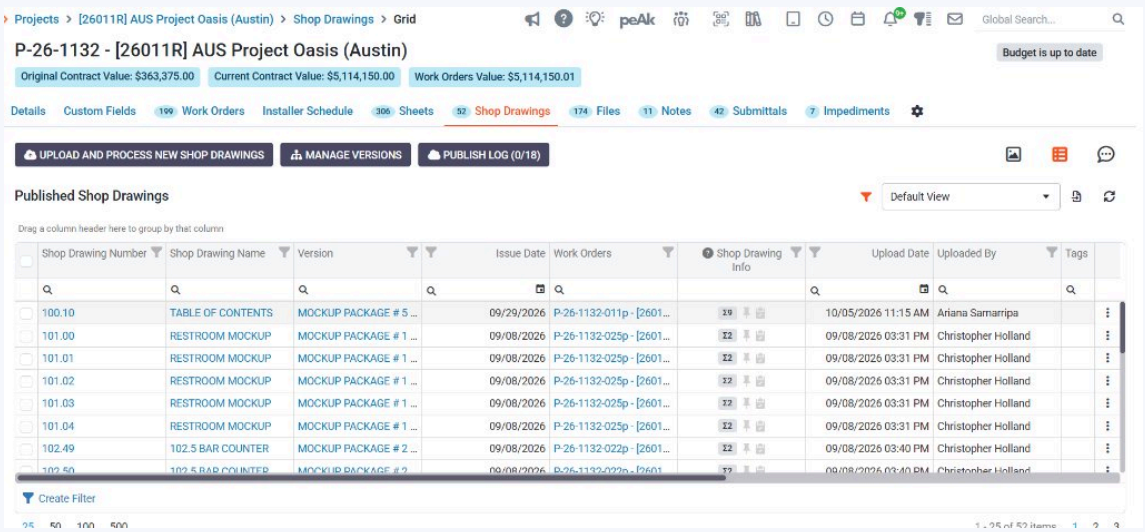
BOTTOM LEFT Complete Revision History Timeline & Version Comparison

- Displays the chronological Revision History list for this specific sheet:
- · 90 IFP (now viewing) · Issued: 07/31/2026
- · Bulletin 01 · Issued: 08/31/2026 (if applicable)
- Clicking any historical revision immediately loads that version's PDF into the right-hand viewer and enables Side-by-Side or Red/Blue Overlay Diff Comparison to spot architectural changes.

RIGHT CANVAS In-Browser Multi-Slide PDF Viewer with Prev/Next Slide Thumbnails

- Streams the PDF directly from AWS S3 / Box using pre-signed URLs so it opens inside the website with zero download delay.
- Slide Count & Carousel Navigation: Shows miniature preview thumbnails of the Previous Slide (<) on the left and Next Slide (>) on the right, plus 'Slide X of Y' counter so users can flip through every slide/page inside a multi-page PDF.
- Interactive Markup Layer: Zoom, pan, calibrate scale measurements, drop Work Order pins, highlight millwork elevations, and add text/redline annotations.

SEC 11 Published Shop Drawings Grid & Mockup Packages (Ref: Image 3)



Reference Image 3: Published Shop Drawings Grid (P-26-1132 AUS Project Oasis) · 52 Shop Drawings Linked to Work Orders

How Shop Drawings Connect Engineering, Work Orders, QC & Submittals

NUMBERING **Architectural Millwork Shop Drawing Hierarchy (Shown in Image 3)**

- 100.10 · TABLE OF CONTENTS (Version: MOCKUP PACKAGE # 5 | Issue Date: 09/29/2026 | Linked WO: P-26-1132-011p | Slide Count: Sigma-9 = 9 slides in PDF).
- 101.00 to 101.04 · RESTROOM MOCKUP (Version: MOCKUP PACKAGE # 1 | Issue Date: 09/08/2026 | Linked WO: P-26-1132-025p | Slide Count: Sigma-2 = 2 slides each).
- 102.49 to 102.50 · 102.5 BAR COUNTER (Version: MOCKUP PACKAGE # 2 | Issue Date: 09/08/2026 | Linked WO: P-26-1132-022p | Slide Count: Sigma-2 = 2 slides each).

WORK ORDER **Bidirectional Linking Between Shop Drawings & Work Orders**

- Every row in the Shop Drawings Grid has a clickable 'Work Orders' column (e.g., P-26-1132-011p, P-26-1132-025p, P-26-1132-022p).
- Clicking the Work Order link jumps straight to that Work Order's production record (showing BOM materials, CNC programs, fabrication hours, and ship date). Conversely, opening a Work Order lists all its attached Shop Drawings.

VERSIONS **Manage Versions (Mockup Packages #1..#5) & Slide Count Badges (Sigma-N)**

- Millwork shop drawings evolve through iterative submittal packages (MOCKUP PACKAGE #1, #2, #5, Rev 0, Rev 1).
- The 'Shop Drawing Info' column displays the total PDF slide/page count badge (Sigma-9, Sigma-2) plus pin and attachment icons. Clicking any Shop Drawing opens the same Split-Pane Multi-Slide PDF Viewer & Revision History shown in Image 4.

PUBLISH LOG **Publish Log (0/18) · Gatekeeping Between Engineering, QC & Client Portal**

- When Christopher Holland or Ariana Samarripa uploads 18 new/revised shop drawings, they sit in 'PUBLISH LOG (0/18)'.
- Once QC Review approves the drawings, the PM opens the Publish Log, selects the approved drawings, and publishes them to the Submittals tab and Client Portal.

SEC 12 Deep Workflow: Study Projects & Work Orders (199 Items)

STUDY PROJ Study Projects Workflow (Pre-Production Technical R&D)

- 1. Create Study Item: Engineer creates a Study Project for high-risk millwork elements (e.g., 'Curved Acoustic Wood Ceiling Mockup', 'Custom Brass Inlay Reception Desk').
- 2. Attach Reference Sheets & 3D Models: Link architectural sheets (G0.02) and upload preliminary CAD/SketchUp/Rhino files to S3/Box.
- 3. Log Engineering Study Hours: Engineers log time against the Study Project budget without cluttering active shop Work Orders.
- 4. Convert to Work Order & Shop Drawing: Once technical feasibility is proven, one click converts the Study Project into an active Work Order and Shop Drawing task.

WORK ORDER Work Orders Lifecycle (From Engineering Release to Shop & Install)

- 1. Work Order Creation: Created during Phase 2 estimating or generated per room/elevation (e.g., P-26-1132-025p Restroom Mockup).
- 2. Engineering & Shop Drawing Link: Shop drawings 101.00 - 101.04 are linked to WO P-26-1132-025p. WO cannot enter 'Production' until linked drawings are 'Approved'.
- 3. Labor Hour Tracking (No Cost Shown to Staff): Tracks Budgeted Hours vs. Actual Logged Hours across Drafting, CNC Cut, Casework Assembly, Finishing, and Packing.
- 4. Shipping & Installer Schedule Sync: When shop marks WO 'Ready to Ship', it automatically populates on the Installer Schedule calendar.

SEC 13 Deep Workflow: Internal QC Review & Formal Submittals (42 Items)

QC REVIEW 4-Stage Internal Quality Control (QC) Gate

- Stage 1 · Automatic Queueing: Every newly uploaded Shop Drawing Version enters the QC Review tab with status 'Pending QC'.
- Stage 2 · Millwork Checklist Inspection: QC Reviewer verifies: (a) Field vs. Arch Dimensions, (b) Veneer Species/Cut & Grain Direction, (c) Edge Banding & Joinery Callouts, (d) Hardware Schedule Match, (e) Wall Blocking & ADA Clearances.
- Stage 3 · In-Browser PDF Redlining: QC draws revision clouds, arrows, and notes directly onto specific slides of the PDF.
- Stage 4 · Decision: If 'Rejected', returns to Engineer for revision. If 'Approved', unlocks the drawing in the Publish Log.

SUBMITTALS 5-Stage Architect / Client Submittal Workflow

- Stage 1 · Package Assembly: PM creates Submittal #012 (Spec Section 06 41 00 Custom Casework) and attaches QC-approved Shop Drawings (101.00-101.04), hardware cutsheets, and finish sample photos.
- Stage 2 · Transmittal PDF Generation: System auto-generates a branded Submittal Cover Sheet & Transmittal Log PDF stored in S3/Box.
- Stage 3 · Client Portal Notification: Client/Architect receives email + portal alert; Ball-in-Court switches to 'Client / Architect' with contractual due date timer.
- Stage 4 · Client Markup & Disposition: Architect reviews slides in-browser and stamps: [Approved], [Approved as Noted], or [Revise & Resubmit].
- Stage 5 · Revision Loop (REVX) or Fabrication Release: If Revise & Resubmit, creates Submittal Rev 1. If Approved, releases linked Work Orders to the shop floor.

SEC 14 AWS S3 / Box Cloud Storage & In-Browser PDF Rendering Pipeline

1. UPLOAD Direct Pre-Signed Upload (Client & Internal Staff)

- When a Client, Engineer, or PM selects a file (PDF, DWG, DXF, Revit, Image, Spec Doc), the browser requests a time-limited Pre-Signed PUT URL from the QuadeConnect API.
- The file uploads directly to the project's isolated S3 bucket path (/projects/{proj_no}/{category}/{version}/{filename}) or Box Enterprise folder.

2. PROCESS Automated PDF Slide Counter, Thumbnail Generator & Title-Block OCR

- An S3 Event / Webhook triggers the Document Processing Worker (PyMuPDF + OCR):
- · Counts exact number of slides/pages in the PDF and writes `page_count` (displayed as Sigma-9, Sigma-4, Sigma-3, Sigma-2 in the UI).
- · Renders WebP thumbnails for every individual slide so the Sheet/Shop Drawing viewer can show instant left/right slide previews.
- · Extracts Sheet/Drawing Number and Title from the PDF title block.

3. STREAM In-Browser Viewing & Non-Destructive Annotation Layer

- Clicking any file, Sheet, or Shop Drawing generates a short-lived Pre-Signed GET URL streamed directly into the embedded PDF.js / Box Preview canvas inside the website.
- Redline markups, pins, and QC stamps are saved as structured JSON vectors in PostgreSQL linked to `(document_version_id, slide_number)`, keeping the original PDF immutable.

SEC 15 Core Relational Database Schema (PostgreSQL Entities)

projects	id, proj_no (P-26-1137), code (26011R), name, customer_id, status, kanban_stage, original_contract_val*, current_contract_val*, wo_total_val*, billed_val*, billed_pct*,
project_assignments	id, project_id, user_id, role_in_project (PM, ENGINEER, QC, OPS_HEAD, INSTALLER, CLIENT), budgeted_hours, logged_hours
user_pay_rates	id, user_id, trade_role, hourly_rate_usd* [*Strictly Admin-only table for computing Labor Cost = logged_hours x hourly_rate_usd]
study_projects	id, project_id, study_no, title, description, status, assigned_engineer_id, budgeted_hours, logged_hours, converted_wo_id
work_orders	id, project_id, wo_number (P-26-1132-011p), name, status, wo_value*, budgeted_hours, actual_hours, ship_date, install_start_date
sheets	id, project_id, sheet_number (G0.02), sheet_name, current_version_id, is_published, tags[]
sheet_versions	id, sheet_id, version_set_name (90 IFP, Bulletin 01), issue_date, slide_count (Sigma-N), s3_key, thumbnails_json, uploaded_by, uploaded_at
shop_drawings	id, project_id, work_order_id, drawing_number (101.00), drawing_name (RESTROOM MOCKUP), current_version_id, qc_status, is_published
shop_drawing_versions	id, shop_drawing_id, version_name (MOCKUP PACKAGE #1), issue_date, slide_count (Sigma-N), s3_key, thumbnails_json, uploaded_by
qc_reviews & markups	id, target_type (SHEET/SHOP_DWG), version_id, slide_number, reviewer_id, status, checklist_json, annotation_vectors_json, created_at
submittals	id, project_id, submittal_no, spec_section, title, revision_no, status, ball_in_court, linked_drawing_version_ids[], transmittal_s3_key

SEC 16 Master Feature Comparison Matrix (ClickUp vs. Innergy vs. QuadeConnect)

Feature / Capability	ClickUp	Innergy ERP	QuadeConnect PM (Target)
Master Projects Grid + Group By + Custom Filters	Yes (Generic)	Yes (Millwork)	**YES · Full Grid with Footer Sums & Saved Views**
Projects Kanban Board (#/projects/kanban)	Yes	Yes	**YES · Millwork Lifecycle Columns + Drag & Drop**
Gantt Timeline & Installer Schedule Calendar	Yes	Partial	**YES · Integrated Shop + Field Installer Schedule**
Contract, Work Order Value & Billed % Tracking	No	Yes	**YES · Full Portfolio & Project Financial Tracking**
Role-Based Cost Hiding (Admin=$ vs Eng/Ops=No $)	Limited	Partial	**YES · Strict API-Level Cost Masking & Project Sc**
Hours Pay Plan (Role $/hr + Labor Hours Budget)	Basic Time	Yes	**YES · Trade Rate Cards (Admin) + Hours Tracking**
Study Projects (Pre-Production Millwork R&D)	No	Custom	**YES · Dedicated Study Projects Tab -> Converts t**
Work Orders Module (199+ WOs per Project)	Tasks Only	Yes	**YES · Full Work Orders Linked to Shop Drawings**
Architectural Sheets Grid + Version Sets (90 IFP)	No	Yes	**YES · Upload, OCR Title Block, Version Sets & Pu**
Shop Drawings Grid + Mockup Packages (#1..#5)	No	Yes	**YES · Linked to Work Orders with Mockup Versio**
PDF Slide Count Badges (Sigma-9, Sigma-4) & Previews	No	Yes	**YES · Auto Slide Count + Split-Pane Slide Carous**
Revision History Panel + Side-by-Side Diff Viewer	Basic Proofing	Yes	**YES · Full Historical Version Stack & Overlay Diff**
Internal QC Review Gate + Millwork Checklists	No	Partial	**YES · Dedicated QC Tab + Redline PDF Annotatio**
Formal Submittals Workflow + Transmittal Logs	No	Yes	**YES · Submittal Packages, Ball-in-Court & REVX**
AWS S3 / Box Direct Cloud Storage + Client Portal	Attachments	Cloud	**YES · Pre-Signed S3/Box Streaming + Client Wor**

SEC 17 Recommended Step-by-Step Build Order for Development

Sprint 1 · Core Foundation, RBAC & Master Projects Hub (Grid + Kanban):
Build PostgreSQL schema, JWT Auth + Role-Based Cost-Masking Middleware, Master Projects Grid (Image 1) with portfolio footer totals, and Projects Kanban Board (#/projects/kanban).

Sprint 2 · Inside-Project Shell, Details, Custom Fields, Study Projects & Work Orders:
Build the persistent Project Header, 11-tab navigation bar, Hours Pay Plan engine, Study Projects tracker, and Work Orders module.

Sprint 3 · AWS S3 / Box Integration, Sheets & Shop Drawings with Multi-Slide PDF Viewer:
Integrate S3/Box pre-signed uploads, PyMuPDF background worker (computing Sigma-N slide counts & slide thumbnails), Sheets Grid (Image 5), Shop Drawings Grid (Image 3), and Split-Pane Revision History Viewer (Image 4).

Sprint 4 · QC Review Gate, Submittals, Client Workspace & Revision Loop (REVX):
Build in-browser PDF redline annotations, QC Checklist queue, Publish Log gatekeeper, Submittal transmittal generator, and restricted Client Portal.